

Assignment 5: Non-functional Testing

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Repository Link:

<https://github.com/mirNNes/bsv-edutask>

1. Qualities

1.1 Why Define Qualities

Non-functional qualities must be explicitly defined before testing because they are abstract. A clear definition makes the quality measurable and allows us to determine whether the system satisfies it.

1.2 Selected Qualities

Accessibility is the degree to which a system can be used by people with different abilities.

Maintainability is the degree to which a system can be modified or improved with low effort.

Reliability is the degree to which a system performs correctly under given conditions.

1.3 Test Techniques

Accessibility: Test using manual inspection and accessibility tools (e.g., checking keyboard navigation and readability).

Maintainability: Test using static code review (checking structure, readability, and modularity).

Reliability: Test by repeatedly executing user scenarios and observing consistent behavior.

2. Static Testing

2.1 Static vs Dynamic Testing

Static testing means examining the system without executing it, for example by reviewing source code. Dynamic testing means executing the system and observing its behavior

2.2 Code Review and Extensibility

A static code review was performed to evaluate the extensibility of the EduTask system for supporting Medium articles.

The system is strongly coupled to YouTube videos:

The frontend requires a YouTube URL (TaskCreator.js)

Tasks are displayed using YouTube thumbnails (TaskView.js, TaskDetail.js)

The converter assumes a video object with a url (Converter.js)

The backend shows the same design:

A video object is always created from data['url'] (TaskController)

Tasks always contain a video field

Because of this, adding Medium articles would require changes in multiple parts of the system, including frontend, backend, and data model.

Therefore, the system has limited extensibility. It can be extended, but not with minimal changes.